GROWTH HORMONE & ANABOLIC SIGNALING

Tesamorelin — the full review

GROWTH HORMONE-RELEASING HORMONE ANALOGUE

This is the deeper look behind the short version — the two large trials that supported approval, why tesamorelin's compounding status is actually stricter than Sermorelin's rather than looser, what the smaller non-HIV and cognition trials found, and what your consultation covers.

What it is

Tesamorelin is a lab-made copy of a 44-amino-acid natural brain hormone, growth hormonereleasing hormone (GHRH), with one chemical addition that slows how quickly the body breaks it down. Like Sermorelin and CJC-1295 in this catalog, it prompts the pituitary gland to release the body's own growth hormone. What sets it apart from both: it is the one FDA-approved medicine in this specific family, marketed as Egrifta, for a single, narrow use — reducing excess abdominal fat in HIV-associated lipodystrophy.

Because tesamorelin keeps nearly the entire 44-amino-acid GHRH sequence — longer than Sermorelin's shortened 29-amino-acid fragment or CJC-1295's similar-length backbone — it crossed a regulatory threshold in 2020 that reclassified it as a licensed biologic rather than an ordinary drug. That reclassification matters more than it sounds: it changes which compounding rules can legally apply to it, in a stricter direction, not a looser one.

What people use it for

USED FOR	WHAT THAT MEANS
Excess visceral fat in HIV-associated lipodystrophy	Roughly 800 people studied across two Phase 3 trials; visceral fat down 11-15% versus placebo, subcutaneous fat unchanged.

USED FOR	WHAT THAT MEANS
Liver fat in HIV patients with fatty liver disease	A dedicated trial found reduced liver fat and slowed fibrosis progression — real evidence, but still within the HIV population and not an approved use.
General visceral fat reduction in adults without HIV	One 12-month trial in 60 people with reduced growth hormone secretion found a benefit; never replicated and not the same population as general obesity.
Cognitive function	One 152-person trial found improved cognition scores over 20 weeks, alongside a rise in fasting insulin in a subgroup; a single, unreplicated finding.
General weight loss or body composition	The FDA label explicitly states tesamorelin is not intended for general weight-loss management; trials found no change in subcutaneous fat.

How it works

Tesamorelin binds the GHRH receptor on pituitary cells, triggering a pulse of the body's own growth hormone, which prompts the liver to produce IGF-1 — roughly doubling it in the pivotal trials. Growth hormone preferentially drives fat loss from visceral fat, the more metabolically active fat around the organs, rather than subcutaneous fat just under the skin, which is the basis for its approved, narrowly targeted use and for the label's explicit statement that it is not intended for general weight loss.

Like other compounds in this family, it only works if the pituitary gland still has growth hormone available to release — it has nothing to stimulate if the pituitary itself is damaged, missing, or unresponsive.

What the research actually looked at

The FDA approval rests on real, sizable Phase 3 evidence: two randomized, placebo-controlled trials in HIV-associated lipodystrophy, together studying roughly 800 people, over 26 weeks (one extended to 52 weeks). Visceral fat fell by roughly 11 to 15 percent more than placebo, while subcutaneous fat was essentially unchanged — a considerably larger and more rigorous evidence base than most compounds in this category, though FDA's own reviewers noted that visceral-fat reduction is a validated but surrogate measure, and the exact clinical benefit of that degree of reduction is less certain.

Evidence outside the approved population is real but much thinner. A 12-month trial in 60 nonHIV obese adults with reduced growth hormone secretion found a similar visceral-fat benefit, but it has never been replicated and applies to a pre-selected subgroup, not general obesity. A separate 152-person trial in older adults, including some with mild cognitive impairment, found improved composite cognition and executive function scores over 20 weeks — a genuine but single, unreplicated finding that also showed a rise in fasting insulin in the cognitively impaired subgroup.

Because tesamorelin is now a licensed biologic rather than an ordinary unapproved drug, it does not appear on FDA's 503A or 503B compounding bulk-substance lists at all — not because it was reviewed and cleared, but because those lists govern drugs, and current FDA guidance states that licensed biologics are not eligible for the standard compounding exemptions available to unapproved peptides. Compounded "tesamorelin" sold for general anti-aging or weight-loss use sits in a more legally uncertain position than compounds like Sermorelin, not a more permissive one.

Common misunderstandings

“FDA-approved means it's endorsed for weight loss generally.” The approval is narrow — HIV-associated lipodystrophy only — and the label explicitly excludes general weight-loss management.

“Compounded tesamorelin is the same kind of gray area as compounded Sermorelin.” It is not, and it is arguably stricter. Because tesamorelin became a licensed biologic in 2020, current FDA guidance states such products are not eligible for the standard compounding exemptions at all.

“The cognition study proves growth-hormone peptides sharpen the mind.” That trial is tesamorelin's own data, from a single 20-week study using surrogate cognitive measures, never replicated, and it showed a meaningful rise in fasting insulin in part of the study group.

“It works like a GLP-1 drug for fat loss.” Trials show a selective effect on visceral fat specifically, with no change in subcutaneous fat — a different pattern than the broad weight loss seen with GLP-1 medicines, and the reason the label excludes general weight-loss use.

Who it is not for

Anyone with disruption of the pituitary gland or its signaling — from prior pituitary surgery, tumor, head irradiation, or head trauma.

Anyone with active cancer or a cancer history — the label carries a warning about increased neoplasm risk observed in treated patients.

Pregnant patients, or anyone with a known hypersensitivity to the medicine.

Anyone with diabetes or unstable blood sugar, without discussing it first — glucose intolerance is a labeled risk.

Anyone expecting a general weight-loss effect — the label explicitly excludes this use, and trials found no change in subcutaneous fat. Competitive athletes: Tesamorelin is named specifically on the WADA Prohibited List, in the same growth-hormone-releasing-hormone category as CJC-1295 and Sermorelin; it is prohibited at all times, in and out of competition, including use of the approved product without a documented therapeutic use exemption.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller. That designation carries specific, enforceable obligations:

A prescription in your name. Compounded from a prescription for one identified patient, after an evaluation.

Licensed and inspected. A state licence, subject to inspection by the state board of pharmacy.

Ingredients that can be traced. A certificate of analysis from an FDA-registered facility, back to a named source and lot.

Sterile preparation to a published standard. USP standards.

A paper trail. A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

An approved medicine exists, for one narrow population. Roughly 800 people studied across two Phase 3 trials supporting approval for excess visceral fat in HIV-associated lipodystrophy specifically — not general weight loss. Compounded tesamorelin sits on stricter legal footing than it might appear. Since becoming a licensed biologic in 2020, current FDA guidance holds that it is not eligible for the standard compounding exemptions available to unapproved peptides — a different, tighter regulatory posture than most compounds in this catalog. Ask specifically whether what's being offered is the approved product or a compounded preparation, and for what population it was actually studied.

Clinical detail

A 44-amino-acid stabilized GHRH analogue; FDA-approved as Egrifta/Egrifta SV/Egrifta WR (NDA 022505, approved November 2010) for excess abdominal fat in HIV-associated lipodystrophy. Reclassified from an approved drug to a licensed biologic on March 23, 2020, under the biologics-deeming provision, because it exceeds the 40-amino-acid length threshold that distinguishes a "protein" biologic from an ordinary drug. Current FDA guidance states licensed biologics are not eligible for the standard 503A/503B compounding exemptions — consistent with tesamorelin's absence from both bulk-substance lists. Pivotal approval trials: two randomized, placebo-controlled studies (n=412 and n=404) in HIV-associated lipodystrophy, 26 weeks each, one extended to 52 weeks. Label explicitly states tesamorelin is not intended for general weight-loss management, and long-term cardiovascular safety has not been established.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment. Tesamorelin is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality. Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose. Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.

IMPORTANT SAFETY INFORMATION

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